

Optimal Bandwidth for Multitaper Spectrum Estimation - Supplementary Material

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This supplement serves to (i) motivate the use of multitaper spectrum analysis over single-taper estimates, (ii) summarize the jackknife method for multitaper variance estimation, described in the literature and referenced in the body of the text, (iii) show the consequences of poor multitaper bandwidth choice on the final estimate, and (iv) provide a derivation for the optimization of MSE in our motivating example.

IX. NOTE ON ASYMPTOTIC UNBIASEDNESS OF THE PERIODOGRAM

Schuster's 1898 periodogram estimator for the spectrum of a stationary process $x(t)$,

$$P(f) = \left| \sum_{n=0}^{N-1} x(t) e^{-i2\pi f t} \right|^2, \quad (19)$$

is an unbiased estimator of the spectrum as $N \rightarrow \infty$ at the Fourier frequencies. This statement has justified the use of the periodogram (or smoothed periodogram) for scientific inference when large samples are available, but we caution that this statement is grossly misleading because convergence can be too slow for practical purposes.

In general, processes that have large spectral range will have poor periodogram and direct estimates since high sidelobes of the periodogram spectral window (Fejér kernel) cannot resolve such a large range. Consider the example in [3, Fig. 1(a)], in which $N = 1,000$ samples of barometric pressure data are used to compute a periodogram estimate that differs from the multitaper with $NW = 8$ by a factor of at least 1,000 across most of the frequency range, and by 10^7 in higher frequencies. Since the multitaper estimator has leakage bounded above (see (6)) by 8.3×10^{-17} , such a discrepancy shows that the sample size $N = 1,000$ is not large enough to use the periodogram for scientific inference. Furthermore, the estimate would not benefit from smoothing. If a multitaper estimate had not been computed on these data, it would not be known that these data have such a large spectral range. In fact, any effort to fit a parametric model to these data based on a periodogram [21, 15] or a Bartlett covariance would result in more high-frequency power in the model than actually exists.

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Although asymptotic unbiasedness of periodograms and single-tapered estimates is important, this fact is of little use when doing analyses on finite samples, since no sample size is large enough to sufficiently reduce the periodogram bias. More examples of bias in periodogram and direct estimates can be found in [5, p. 227].

X. JACKKNIFING MULTITAPER SPECTRUM ESTIMATES

In this section, we give an overview of jackknifing multitaper spectrum estimates in order to obtain variance estimates and confidence intervals. The spectra, $\hat{S}^{(k)}(f)$, produced as intermediates of the multitaper computation, are approximately independent, by virtue of having been computed with orthogonal tapers, and hence can be used to produce an estimate of variance. The *delete one jackknife* method [17, 18, 11, 22], described here gives an estimate having less bias than one computed directly as a sample variance of the individual spectra [19] and is statistically efficient for computation on a small number of independent samples.

Beginning with the spectra, $\hat{S}^{(k)}(f)$ from (1) for $k = 0, \dots, K-1$, one computes *delete one* estimates of the log spectrum

$$\ln \hat{S}_{\setminus j}(f) = \ln \left[\frac{1}{K-1} \sum_{\substack{k=0 \\ k \neq j}}^{K-1} \hat{S}^{(k)}(f) \right] \quad (20)$$

$$= \ln \left[\frac{1}{K-1} (K \bar{S}(f) - \hat{S}^{(j)}(f)) \right]. \quad (21)$$

The average of the delete one estimates is denoted

$$\ln \hat{S}_{\bullet}(f) = \frac{1}{K} \sum_{j=0}^{K-1} \ln \hat{S}_{\setminus j}(f). \quad (22)$$

The jackknife variance estimate is computed as

$$\hat{\sigma}_J^2(f) = \frac{(K-1)^2}{K(K-\frac{1}{2})} \sum_{j=0}^{K-1} [\ln \hat{S}_{\setminus j}(f) - \ln \hat{S}_{\bullet}(f)]^2. \quad (23)$$

The expected value of this last is

$$\mathbf{E}\{\hat{\sigma}_J^2(f)\} = \frac{(K-1)^3}{K(K-\frac{1}{2})} \left[\frac{2}{(K-2)^2} + \frac{1}{2} \left[\psi' \left(\frac{K-1}{2} \right) - \psi' \left(\frac{K-2}{2} \right) \right] \right]. \quad (24)$$

Where ψ' is the trigamma function,

$$\psi'(t) = \frac{d^2}{dt^2} \ln \Gamma(t); \quad \Gamma(t) = \int_0^\infty x^{t-1} e^{-x} dx. \quad (25)$$

In general, jackknife variance agrees well with its expected value when the data are stationary. One can then construct confidence intervals for the log spectrum according to

$$\bar{S}(f)e^{t_{K-1}(\alpha_0/2)\hat{\sigma}_J} < S(f) \leq \bar{S}(f)e^{t_{K-1}(1-\alpha_0/2)\hat{\sigma}_J}. \quad (26)$$

In the Lorentzian example, the variances follow (24) almost exactly. Examples of jackknife standard deviation are illustrated in the next section. We note that the multitaper method is the only nonparametric method for which one can compute confidence intervals for the spectrum estimate.

XI. MOTIVATION FOR THE OPTIMIZATION OF MULTITAPER BANDWIDTH

As an example, we use the Lorentzian spectrum (17) with parameters $A = 100$, $\alpha = 0.01$, $f_0 = 0.2$, and $N = 100$ introduced in the main paper. Simulating this process and estimating the spectrum by using the multitaper method with various choices of bandwidth parameter, we obtain the spectrum estimates shown in Figs. 3 and 4, where the true spectrum is shown in dashed red. Dash-dotted vertical lines show one bandwidth away from the peak and have the same color as the associated spectral estimate. The overall impression is that the shape of the peak reflects the shape of the multitaper spectral window (7), which is square with width $2W$.

In Figs. 3 and 4, two spectrum estimates with different bandwidth choices are shown together with shaded 5% and 95% confidence intervals (26).

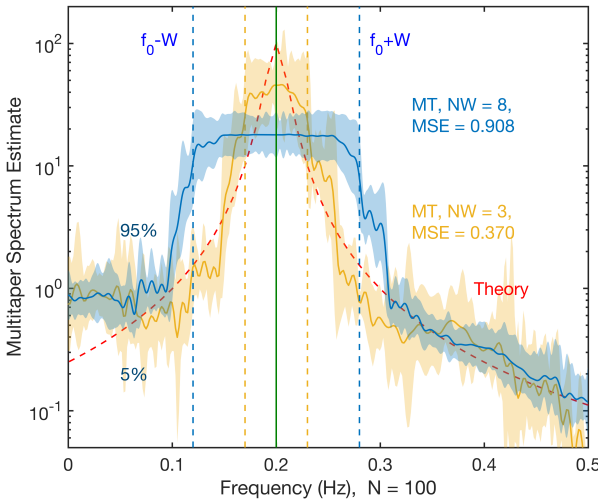


Fig. 3. Two multitaper spectrum estimates computed from simulated Lorentzian data using $NW = 3$ in yellow (lighter shading) and $NW = 8$ in blue (darker shading) along with their jackknife 90% confidence intervals. These are single realizations and have not been averaged. Dot-dashed vertical lines show the bandwidth. Expected jackknife variances for these log spectra are 0.1807 and 0.0645, and MSE are 0.370, 0.908 for $NW = 3$ and for $NW = 8$, respectively.

Figure 3 shows that choosing too large an NW results in a spectrum with large bias. Comparing the time-bandwidth $NW = 8$, blue line with $NW = 3$, yellow line, the latter much better represents the peak; but it is accompanied by larger jackknife variance (0.1807 vs 0.0645, respectively, given

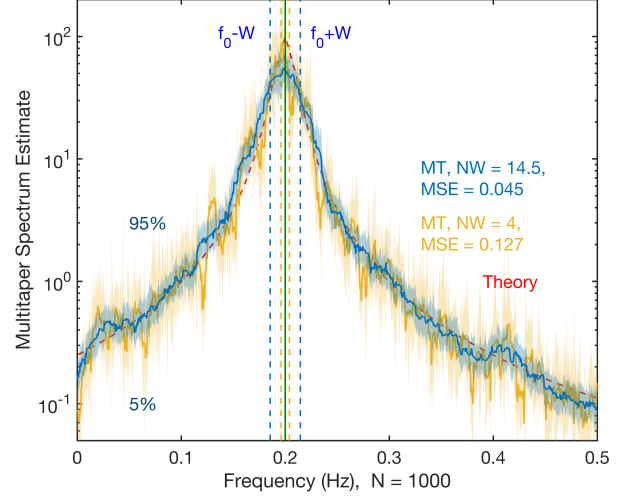


Fig. 4. Two spectrum estimates using $NW = 4$ in yellow (lighter shading) and $NW = 14.5$ in blue (darker shading) along with their jackknife 90% confidence intervals and mean squared errors. Here the sample size has been increased to $N = 1,000$ in comparison with the spectra shown in Fig. 3, while all other parameters remain the same. The expected jackknife variances of these log spectra are 0.1330, and 0.0351, and the mean squared error between the true log spectrum and the two log estimates, integrated over all frequencies, is 0.127 for $NW = 4$ and 0.045 for $NW = 14.5$.

in log-scale units). Here the superior choice is $NW = 3$, since the empirical MSE is reduced by about 40% (0.370 vs 0.908, respectively) on a log scale. Note that with a series of only $N = 100$ data points, no spectrum estimate looks acceptably close to its true value; but from this example it is clear that one dearly pays the price in bias if one chooses too large a bandwidth. The periodogram of this series (not shown) has a range three orders of magnitude larger.

An overly variable estimate results when one chooses a bandwidth that is too small. Using the same Lorentzian parameters as before but increasing the sample size to $N = 1,000$, we show the comparison of time-bandwidth $NW = 4$ with $NW = 14.5$ in Fig. 4. The choice of smaller bandwidth results in larger confidence intervals (expected jackknife variance of 0.1330 and 0.0351 log spectrum units, respectively) wanders farther about the true spectrum than the other and has almost triple the mean squared error (0.127 vs 0.045, respectively). The optimal time-bandwidth product, as determined by our method is $NW = 14.5$. The choice of $NW = 4$, the “default” time-bandwidth product, performs poorly on this example.

In summary, adjustment of the bandwidth parameter allows us to exchange bias for variance, with small bandwidth resulting in small bias but large variance. As these examples show, unwise bandwidth parameter selection can result in a poor spectrum estimate.

XII. LORENTZIAN OPTIMAL BANDWIDTH

In this section, we derive an asymptotic result for optimal bandwidth based on the MSE of the spectrum in terms of the derivatives of the spectrum, and we determine this optimal bandwidth for the toy example when the spectrum is known to be Lorentzian. Note that this result is valid when the MSE

of the spectrum and not its logarithm is used to find optimal bandwidth, the relationship between which is illustrated in Fig. 2.

A. Theoretical Variance of the Multitaper

One can express in numerous ways the variance of the simple multitaper estimate (1) in full generality; but when the spectrum is C^∞ , see [23, Eqs. (4.20, 4.21)], we can write

$$\begin{aligned}\text{Var}\{\bar{S}(f)\} &= \frac{1}{K^2} \sum_{j,k=0}^{K-1} \text{Cov}\{\hat{S}^{(k)}(f), \hat{S}^{(j)}(f)\} \\ &= \frac{1}{K^2} \sum_{p,q=0}^{\infty} D_{p,q} S^{(p)}(f) S^{(q)}(f),\end{aligned}\quad (27)$$

where $S^{(p)}(f)$ is the p th derivative of the true spectrum. The coefficients $D_{p,q}$ are

$$D_{p,q} = \int_{-W}^W \int_{-W}^W \frac{\nu^p}{p!} \frac{f^q}{q!} \left| \sum_k \lambda_k U^{(k)}(\nu) U^{*(k)}(f) \right|^2 d\nu df. \quad (28)$$

From (28), we see that $D_{0,0}$ is K , $D_{p,q} = D_{q,p}$, and $D_{1,0} = 0$ from orthogonality of the Slepian functions and symmetry $U_k(f) = U_k^*(-f)$. Approximating $D_{1,1}$, we obtain [23, Eq. (4.19)]

$$D_{1,1} \approx \frac{4}{15} \cdot \frac{\sqrt{6}}{\pi} \cdot 2NW^3. \quad (29)$$

This results in the following first-order Taylor formulation for multitaper variance,

$$\text{Var}\{\bar{S}(f)\} \approx \frac{1}{2NW} S(f)^2 + \frac{2}{15} \frac{\sqrt{6}}{\pi} \frac{W}{N} S'(f)^2, \quad (30)$$

where $K = 2NW$ is used.

B. Theoretical Expressions for MSE

We can express the MSE of the spectrum in terms of the derivatives of the true spectrum by combining expressions (3) and (30), ignoring broadband bias,

$$\begin{aligned}\mathbf{E}\{(\bar{S}(f) - S(f))^2\} &\approx \frac{W^4 (S''(f))^2}{36} + \frac{S^2(f)}{2NW} \\ &\quad + \frac{2}{15} \frac{\sqrt{6}}{\pi} \frac{W}{N} (S'(f))^2,\end{aligned}\quad (31)$$

where the first term is the squared bias, the last two terms are variance (27), and we have used the first-order Taylor series approximation.

Explicitly, formula (31) gives the MSE at frequency f for a multitaper estimate $\bar{S}(f)$ with bandwidth W estimated from a finite sample of length N , when the true spectrum of the process is $S(f)$. To obtain a single index that measures the error, we simply integrate (31) over $f \in [-1/2, 1/2]$. In the next section we obtain an analytic expression for the integrated MSE for the Lorentzian spectrum. This expression depends on the half-width at half-maximum, α ; the number of samples, N ; and the bandwidth of the estimator, W . We optimize the integrated MSE over the bandwidth.

C. Bandwidth Optimality

To find the optimal bandwidth for the Lorentzian example, we begin by deriving an asymptotic expression for the integrated MSE. Equation (31) contains squared derivatives of the spectrum. Thus, referring to (17) and substituting $f_0 = 0, A = 1$ for simplicity (since the optimal bandwidth will depend only on α and N) one gets

$$\frac{d}{df} L(f) = \frac{-2\alpha^2 f}{(\alpha^2 + f^2)^2} \quad (32)$$

$$\frac{d^2}{df^2} L(f) = \frac{2\alpha^2 (3f^2 - \alpha^2)}{(\alpha^2 + f^2)^3}. \quad (33)$$

The quantity of interest is (31). Integrating the squared quantities over the entire frequency range and retaining only the leading term of a Taylor series in α centered at 0, one gets

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} (L(f))^2 df \asymp \Gamma_0 \alpha \quad (34)$$

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} (L'(f))^2 df \asymp \Gamma_1 \frac{1}{\alpha} \quad (35)$$

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} (L''(f))^2 df \asymp \Gamma_2 \frac{1}{\alpha^3} \quad (36)$$

where Γ_0, Γ_1 , and Γ_2 are positive constants. The integrated MSE then becomes asymptotically

$$\frac{\pi W^4}{48 \alpha^3} + \frac{\pi \alpha}{4 NW} + \frac{\sqrt{6} W}{30 N \alpha}. \quad (37)$$

To find the bandwidth that minimizes integrated MSE, one sets to zero the derivative of the above expression with respect to W , and solves to obtain

$$W_{opt} = c_0 \left(\frac{\alpha^4}{N} \right)^{1/5}, \quad (38)$$

where c_0 is some positive constant. Back-substituting into the MSE expression (37) results in an integrated squared error on the order of

$$c_1 \left(\frac{1}{N^6 \alpha} \right)^{1/5}, \quad (39)$$

where c_1 is some positive constant. As long as $N > \alpha^{-1/6}$ the error is under control, which implies a bandwidth choice of $W < \alpha^{5/6}$, asymptotically.

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